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Enhancing Learning Engagement: A Study on Gamification's Influence on Motivation and Cognitive Load

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Abstract: Research has demonstrated that engagement in any learning environment improves learning and may even boost performance. Consequently, numerous scholars have examined various approaches to raising student engagement in learning. Gamification is one such approach because it is thought to increase motivation and reduce cognitive load to ultimately improve learning outcomes. However, some studies have called into question its benefits, prompting more research to be conducted to properly understand the phenomenon. Hence, the study examined gamification's influence on motivation and cognitive load for enhancing learning engagement. The study was underpinned by the integration of the Attention, Relevance, Confidence, and Satisfaction (ARCS) model, self-determination theory (SDT), and cognitive load theory (CLT) in developing a conceptual framework. A sample of 407 university students who participated in five gamified courses were involved in the study. The data collected through a questionnaire were analyzed using SmartPLS structural equation modeling software. According to the study's findings, both motivation and cognitive load influence engagement in learning, with cognitive load being the stronger influence. The study contributes to our knowledge by elucidating the relationship between gamification and students' cognitive load and motivation to learn, as well as how these elements ultimately engage students in the learning process.

Keywords: gamification; motivation; cognitive load; structural equation model; learning engagement; ARCS; CLT; SDT

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1. Introduction

Engagement in any learning setting enhances learning and may even improve performance, according to research [1–4]. Moreover, Aupperlee's [5] research shows that including students in interactive activities enhances their academic performance. However, it has been a challenge to get students to engage in learning in higher education, and part of this challenge can be linked to the fact that lecturers have not always applied appropriate teaching methods to have the desired effect on students' learning [6]. A number of research studies have confirmed that employing innovative teaching strategies can affect student engagement and, in turn, have a favorable effect on learning outcomes. For instance, it has been discovered that using board games in science classes increases student involvement and comprehension of difficult ideas [7]. This suggests that adopting methods to ensure that students interact with the subject matter to increase learning is just as important as having subject matter experts to facilitate learning. According to Lumpkin, Achen, and Dodd [8], a teacher's mandate is to help students interact with the material in their courses so they can learn. Consequently, numerous scholars have examined various approaches to raising student engagement in learning. Active learning, an instructional approach that encourages students to participate in the learning process through various techniques, is being used. Rather than passively listening to the teacher, the students are involved in the

thinking and reflection process using various activities such as reading, discussions, case studies, group tasks, problem-solving activities, and online learning, using the learning management system and other media. Moreover, researchers in computer science have examined a number of engagement techniques, with the usage of technology-enhanced game features being the most important [9,10].

Gil-Doménech and Berbegal-Mirabent [11] agree that it is important to use innovative pedagogical approaches in higher education to boost student engagement in learning. In the present era, we are more likely to use technology because it increases our productivity, offers high-quality results, or makes us happier. Hence, integrating technology into the classroom is expected to provide high-quality instruction, making learning more interactive and engaging as well as enjoyable. Therefore, it is anticipated that the quality of the teaching and learning processes will be improved with the integration of digital game-based learning in education.

Gamification has been applied in many fields with some success, including health [12], business [13], sports [14], management [15], and education [16]. Gamification, or the use of game elements in non-game contexts, is gaining traction in the educational field as a way to boost student engagement, motivation, and performance [17–19]. In education, gamification promotes learning in a way that is memorable and engaging for students, which results in authentic learning and can develop creative thinking [20]. In a very recent study, Lee [21] contends that gamification can enhance students' motivation, attitudes, and interest in studying, as well as enhance their teamwork, creativity, and communication skills. Aupperlee's [5] study has raised awareness of the need for effective teaching and learning to be hands-on, heart-on, and minds-on; applying gamification in education is likely to meet this need.

While the use of game-based ideas in teaching and learning to motivate students, encourage active learning, and solve motivational difficulties has been widely accepted, some research suggests that gamification is ineffective at improving learning outcomes or student engagement [2]. In addition, gamification's critics contend that gamification hinders learning by elevating the stress of competition over necessary levels and ignoring the educational needs of some students. Nevertheless, gamification advocates assert that gamification has the potential to address many of the issues and misconceptions raised in this discussion [22].

There have been several attempts to implement gamification in the classroom. However, Rabah et al. [22] and numerous others noted that gamification is context-sensitive, necessitating additional research to fully comprehend the phenomenon and confirm its benefits. Even though many studies on gamification in teaching and learning have been conducted, few empirical studies have been carried out in higher education in different subject specializations [23].

Furthermore, it is necessary to fully understand the impact of gamification on students' cognitive load and motivation to learn, as previous studies have only partially shed light on these aspects. The effect of gamification may vary greatly throughout educational levels, subjects, cultural contexts, and age groups due to its context-sensitivity. Therefore, in order to develop successful, flexible instructional strategies that can be broadly used, it is imperative to understand how gamification affects cognitive load and motivation in varied learning situations. While motivation and cognitive load are frequently examined independently, more research is required to understand how gamification affects cognitive load and motivation together. Gaining an understanding of this relationship can assist teachers in striking a balance between cognitive demands and motivational techniques, resulting in better-integrated teaching methods that promote deeper and more engaged learning. With this information, educators and policymakers may create educational solutions that reduce cognitive load and increase student motivation and create an engaging learning environment. Learners who are motivated are more likely to engage in active participation, overcome challenges, and produce superior learning results. Additionally, using a system or instructional materials that reduce cognitive load enables students to

process information more quickly, thus improving understanding, thinking clarity, and knowledge retention. Hence, this study seeks to determine how gamification affects students' cognitive load and motivation to learn, which leads to learning engagement. The study is therefore guided by the following question: How can gamification increase motivation and decrease cognitive load to positively impact student engagement and learning in higher education?

The remainder of the article is organized as follows: first, it presents literature relevant to the study; then, it develops hypotheses to answer the question; it then analyzes and discusses the findings. Lastly, implications for practice and concluding remarks are provided.

2. Literature Review

2.1. Attributes of Engagement in Learning

Research has compellingly shown that student engagement in learning is complex and involves elements of behavioral, affective, and cognitive phenomena [8,24,25].

Behavioral engagement concerns conduct, academic achievement, attendance at lectures, involvement in class discussions, and attitude toward difficult work [26–28]. Therefore, teachers must pay attention to the behavioral element of student engagement because it is the one that is measured most frequently in a country [29].

Affective engagement, on the other hand, relates to emotions that students display toward their learning process and surroundings, such as pride, joy, excitement, and curiosity [30]. A student's emotions toward his peers, teachers, education, and school are also considered to be a part of their affective engagement [31]. Schaufeli et al. [32] assert that emotionally engaged students are able to recognize the significance and goal of their educational journey as well as the interactions that lead to improved learning, accomplishment, and fulfillment. D'Errico et al. [33] claim that there is a strong correlation between behavioral engagement and emotions. However, according to Kahu [34], there has not been much research on affective involvement, even though it affects learning.

The cognitive component of engagement concerns the learner's perceptions, beliefs, and mental awareness when processing matters that are significant to them. This covers the amount of time, interest, and constructive attention devoted to organizing and planning learning-related matters [35]. Research indicates that students who are cognitively engaged place a high value on academic work because they are more likely to reflect on the material and attempt to solve problems [34]. Therefore, according to Christenson et al. [25], students who are cognitively engaged perform better, particularly when it comes to assignments involving critical thinking. Gamification is noted to engage students in various ways [36,37].

2.2. Influence of Gamification on Motivation to Learn

Motivation is defined as the psychological component that piques a person's interest in taking action. There are two primary types of motivation, namely, extrinsic and intrinsic. Extrinsic motivation refers to the external benefits or rewards that an individual receives, which stimulate that individual to do something or behave in a particular way. In the context of gamification, typical instances of extrinsic motivation are the points and badges that are given out for finishing tasks successfully. Intrinsic motivation, on the other hand, refers to the satisfaction that comes from finishing the task itself.

Gamification has gained popularity across a range of industries, most notably education, as a tool for motivation and engagement [38]. Students admit to a high level of motivation and interest in learning because gamified activities are competitive [11]. Bouchrika et al. [2] have affirmed in their study that gamification might be a valuable method for motivating students in learning environments and boosting their interaction and engagement. The literature review on gamification has also revealed that many students who feel alienated by traditional teaching and learning methods find some relief from gamification, and that gamification could be a partial solution to the current decline in student motivation and engagement in the educational system [39]. Alsawaier [39] also believes that there is a disconnect between gamification theory and practice. Interestingly,

in a recent comprehensive review of gamification in science education [17], it has been shown that gamification dramatically boosts student motivation and engagement. This review shows that gamification encourages students to learn actively and develops their capacity for scientific thought [17]. In another systematic review study of 18 publications, it was found that gamified instruction in higher education has many advantages, including increased student motivation, engagement, and academic success [4]. These studies, among others, show that gamification is an effective teaching strategy in higher education that can drastically improve student engagement and change the learning process. Montenegro-Rueda et al. [4] further stressed the necessity for more investigation to completely comprehend how gamification affects various academic fields.

2.3. Influence of Gamification on Cognitive Load of a Learner

Cognitive load refers to the degree of difficulty that people have in remembering what they have been taught.

The use of computer games in the classroom has the potential to foster creative thinking, improve performance, and enhance the learning process. This is the result of a study conducted by Papadakis [40], who had students at the University of Crete in Greece utilize computer games in the classroom. Also, immersing students in gamified stories can lessen their cognitive load and improve their understanding of particular subjects, according to Andrew and Carman [41]. In agreement with Andrew and Carman [41], Darejeh [42] showed in his thesis that, in contrast to no narrative and unfamiliar narrative systems, an e-learning system with a well-known story lowers cognitive load. Yukselturk, Altıok, and Başer [43], in addition, suggest incorporating game-based learning strategies into the instruction of foreign languages since they found that there was a significant improvement in cognitive abilities, attitude, and self-efficacy when comparing the average test scores of the experimental group and the control group.

Gamification is becoming more and more popular as a means of enhancing instructional content and, eventually, improving concept retention rates in educational settings [44]. However, Sanchez, Langer, and Kaur [44] contend that there is a lack of theoretical support for the effects of gamification and called for additional study on the novel effects of gamification as well as how individual traits, such as cognition, are influenced by gamification. A systematic literature analysis conducted by Manzano-León et al. [45] also found that gamification can help students memorize concepts and improve their academic performance.

Gamification is one approach that shows promise for increasing participation in cognitive tasks. Although some studies have yielded contradictory results, many others have discovered that gamification enhances performance. Part of the reason for these contradictory findings is that most gamification studies on cognitive load have used careless and inconsistent design techniques [46]. Most of the work carried out on gamification has not distinguished between motivation and cognitive load in a systematic fashion. Thus, further investigation is required to explore how gamification and cognitive processes interact to impact learning engagement. Table 1 shows three separate comprehensive reviews on gamification based on Systematic literature review (SLR) using the PRISMA model (Preferred Reporting Items for Systematic Reviews and Meta-Analysis.

Table 1. A cross-section of reviews on gamification.

Study	Objective	No. of Articles Synthesised	Findings on Motivation	Findings on Cognitive Load
Kalogiannakis et al. [17]	To present the empirical findings of literature on the use of gamification in science education.	24	A general increase in motivation, suggesting the potential benefits of gamification when used in an educational setting.	Not many studies have demonstrated that using gamification to assist in teaching science has improved students' cognitive and higher-order thinking skills.

Table 1. *Cont.*

Study	Objective	No. of Articles Synthesised	Findings on Motivation	Findings on Cognitive Load
Manzano-León et al. [45]	To review the data that have been collected over the past five years about the effects of educational gamification on student motivation and academic achievement.	14	Motivation was found to be the most studied variable in gamification	No specific finding was stated except that an improvement in student academic achievement was observed.
Montenegro-Rueda et al. [4]	To compile data on gamification research conducted in higher education and evaluate its effects on the learning of university students.	20	The motivation generated by students is the most important aspect of gamification, according to the general findings.	No specific finding was stated.

It could be gathered from Table 1 that further studies need to be conducted on the impact of gamification on the cognitive load of a learner.

3. Related Theoretical Models

The study is underpinned by integrating three theories relevant to motivation, behavioral change, and cognitive load, which are discussed next.

3.1. Motivation-Related Models

The Self-determination theory and the Attention, Relevance, Confidence, and Satisfaction (ARCS) motivational model are pertinent theories related to motivation and behavior toward use of technology [47].

3.1.1. Self-Determination Theory of Motivation

The self-determination theory (SDT) [48] helps to comprehend human motivation. Figure 1 depicts the self-determination theory as espoused by Ryan and Deci [49].

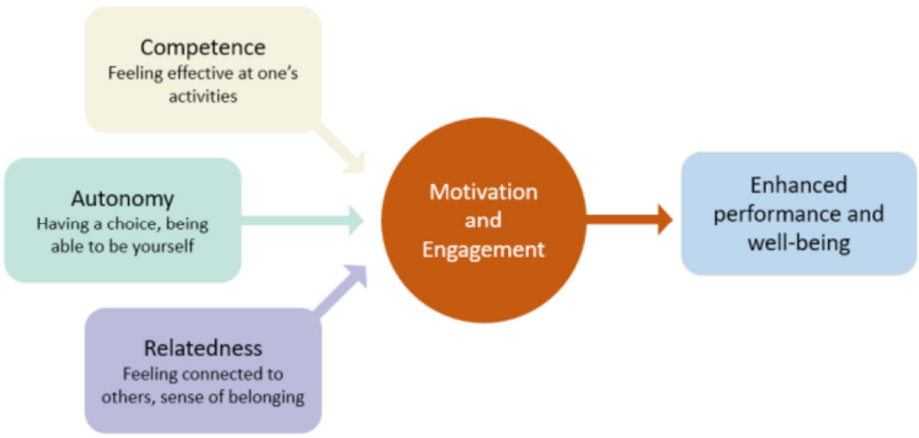

Figure 1. Self-determination theory (Source: [49]).

Self-motivation is determined by the degree to which three basic needs, autonomy, competence, and relatedness, are satisfied in each individual [48]. Being autonomous means having the freedom to choose without being forced to make a decision or having control over one's activities. A sense of mastery over a task or efficacy in completing an activity is indicative of competence. Relatedness refers to feeling connected to and accepted by others. It can be inferred from their theory that extrinsic motivators can be used to

encourage people to try a task, and once they start, they can naturally develop a sense of satisfaction. In other words, humans are driven to do things that satisfy them, not by external factors like rewards or punishment avoidance. In the same vein, learners can be introduced to a course or learn the subject matter through gamification, which lends itself to be an intrinsic motivation for influencing learning. According to Gagné et al. [50], satisfying the requirements of competence, autonomy, and relatedness, which are linked to both intrinsic and extrinsic types of motivation, is essential for human development and engagement.

Ryan and Deci [48] further suggest that to use motivation effectively within the context in which it is employed, it is necessary to identify the individual profiles and motivational factors, a claim that Ruhi [51] concurs with. According to Dichev et al. [52], the SDT provides a foundation for comprehending how motivation affects gamification [53].

3.1.2. Keller's ARCS Motivational Model

John Keller's [54] Attention, Relevance, Confidence and Satisfaction (ARCS) motivational model offers the necessary features to successfully motivate students. These four characteristics are frequently cited as critical foundations for learning performance [55,56].

The diagrammatic representation of the effects of gamification on each motivational attribute of ARCS is shown in Figure 2.

Figure 2. Gamification and Keller's motivational model (Source: [57]).

The term "attention" suggests that the information must captivate readers in a way that will hold their interest. This is accomplished in a number of ways, including by promoting role playing, employing visual aids, and encouraging involvement. For learners to recognize a course's value both now and in the future, its contents must be relevant to them. The key component of confidence is to allow students to complete tasks that are moderately challenging to maintain their interest. The satisfaction component makes sure that students are happy with the course and their performance. This can be achieved by providing feedback and incentives that are both realistic and engaging, as well as by removing any possibility of failing.

The identification of the prerequisites for putting gamified systems into place is influenced by the use of Keller's motivational model [56]. In order to comprehend how the many components of motivation affect students' engagement in the learning process, it is necessary to take this theory into account in this study.

3.2. Cognitive Load Theory

The cognitive load theory (CLT), which was created by Sweller and Chandler, helps instructional designers create lessons that minimize the cognitive load, that is, the amount of mental energy that students must use. The theory holds the notion that an instructional

design should minimize cognitive load in order to improve learning [58]. It makes a further distinction among intrinsic, extraneous, and germane cognitive loads, as shown in Figure 3.

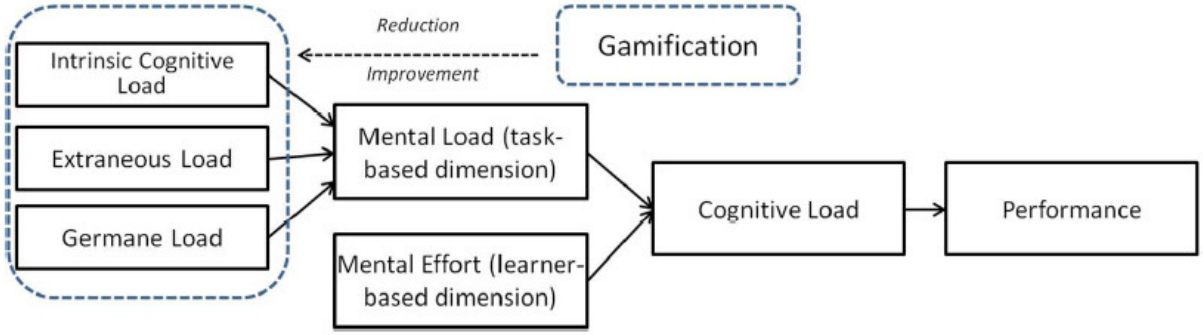

Figure 3. Gamification and cognitive load theory (Source: [58]).

Intrinsic cognitive load refers to the degree of difficulty that is inherent in a given topic. Extraneous cognitive load makes up the excess reflections, which needs to be minimized to lower cognitive load. Reducing the amount of effort needed to develop domain-specific knowledge is the germane cognitive load.

In order to improve performance and eventually increase engagement, the cognitive load theory has been successfully employed in the construction of learning materials [59]. The cognitive load theory has been examined in this study in order to better understand how course module design can lessen the mental work required to comprehend the material.

4. The Adapted Research Model

Since research on motivation's impact on learning has mainly looked at either ARCS or SDT, an integrated model developed by Baah, Govender, and Rontala [60] was utilized to measure motivation.

It is claimed that giving a student the impression that they have autonomy, that is, they are in charge of their own actions, will help them feel more confident in the course. Therefore, the competence and autonomy of SDT can both be explained by the confidence of ARCS [60].

The ARCS model's attention, relevance, confidence, and satisfaction are significant antecedents of motivation, and the self-determination theory's (SDT) relatedness, autonomy, and competence are significant contributory factors of motivation influencing the conceptual framework. They all contribute to our understanding of motivation, a crucial precondition for engagement. As a result, motivation in the conceptual framework includes all of the ARCS model's elements, including the relatedness and implicit competence and autonomy constructs of SDT, which are convergent to the confidence of ARCS as previously mentioned. Thus, motivation is dependent on Attention, Relevance, Confidence, Satisfaction, and Relatedness (ARCSaR) [60]. Conversely, Cognitive load theory affects performance, which may be due to increased engagement. Thus, it must be considered in the conceptual framework as well.

Using multiple theories to better understand the complexity of human behavior [61], the study is supported by the conceptual framework as presented in Figure 4, and is derived from the theories described above.

Figure 4 opines that a gamified course module (gamification) is expected to motivate students to learn and also reduce their cognitive load to ultimately engage them in learning. The dotted line suggests that the gamified course is encapsulated in other factors. In other words, the gamified course is implemented using technology and facilitation process or delivery methods. The technology comprises tools and platforms such as learning management systems and incorporates game elements such as Points, Badges and Leaderboards.

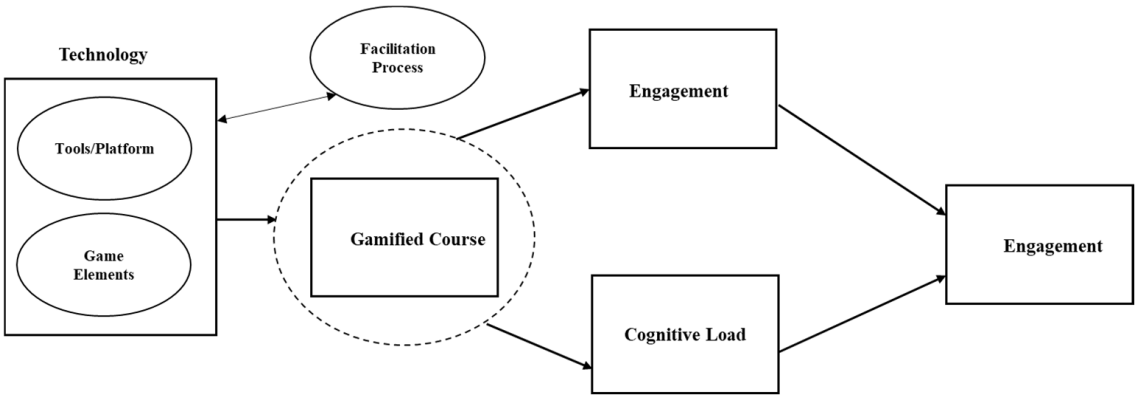

Figure 4. Conceptual framework: Gamification in teaching and learning.

5. Hypotheses Development

5.1. The Influence of Motivation on Engagement

As indicated earlier, motivation is the psychological element that inspires a person to carry out an activity or behave in a particular way. Researchers such as Caserman et al. [38], Gil-Doménech and Berbegal-Mirabent [11], and Bouchrika et al. [2] have indicated that gamification results in motivating students to interact with learning material and engage in learning. Hence, the following was hypothesized:

H1. *Through gamification, motivation (M) significantly relates to engagement in learning (E).*

5.2. The Influence of Cognitive Load on Engagement

According to Manzano-León et al. [45], gamification can help students to memorize concepts and improve their academic performance, as indicated earlier. Darejeh's [42] thesis has also shown that gamification lowers cognitive load. Consequently, we hypothesized the following:

H2. *Through gamification, cognitive load (C) influences engagement in learning (E).*

6. Methodology

6.1. Approach

This exploratory study, which is part of a larger PhD study [62] that develops a framework for gamification, was carried out using a quantitative methodology. One possible benefit of employing quantitative research techniques is that the data may be less biased, enabling the researcher to independently confirm that the participants match the study's requirements without disclosing any additional personal information.

6.2. Population and Sample

There were 407 participants in the target group, consisting of 195 first-year students, 88 second-year students, 60 third-year students, and 64 fourth-year students.

6.3. Intervention

Five courses (modules) from the Information Technology (IT) department were gamified by three (3) different lecturers who were anonymously called Lecturer 1, Lecturer 2, and Lecturer 3. Lecturer 1 gamified three modules, namely, IT and Computer Fundamentals, which is a generic module for all the undergraduate first-year students comprising 195 students (Day and Weekend students); Information Systems Management, a third-year module for 13 IT weekend students; and IT Project Management, a fourth-year module for 64 IT Day students. Lecturer 2 gamified one module, Systems Analysis and Design, which was a second-year module for 88 day-students. Lecturer 3 gamified one module, Information Security, which is a module offered for 47 third-year day students. The five

modules were chosen at random, but care was taken to include both day and weekend course schedules and four levels from first to fourth-year modules. Table 2 is a summary of the modules and students

Table 2. Summary of the demographics, gamified modules, and participants.

Level	Module	No. of Students	No. of Respondents	Response Rate
100	IT and Computer Fundamentals	195	186	95%
200	Systems Analysis and Design	88	83	94%
300	Information Security	47	45	96%
	Information Systems Management	13	11	85%
400	IT Project Management	64	62	97%
	Totals	**407**	**387**	**95%**

6.4. Data Collection Instrument and Measures

The data collection instruments were adapted from previous research. The constructs—motivation, cognitive load, and engagement—were measured using questionnaires adapted from Keller [63], Leppink et al. [64], and Hart, Stewart, and Jimerson [65], respectively. A 5-point Likert scale was used, with 1 representing "strongly disagree" and increasing to 5 representing "strongly agree". Each respondent, on average, spent at least 20 min to fill the questionnaire. The questionnaires were administered online through the institution's LMS after the respondents had taken a semester course, as indicated in Table 1. To enhance the rate of response, students were regularly reminded to complete the online questionnaires. Of the 407 participants, 387, representing 95%, responded (see Table 1).

6.5. Ethical Issues

The participants were informed that their participation was voluntary and that their privacy would be protected prior to the questionnaires being given to them. Moreover, before sending out the online questionnaire, the respondents were made aware of the study's objectives.

6.6. Data Analysis

The partial least square structural equation model (PLS-SEM), with the assistance of SmartPLS 4 software, was used to analyze the data. PLS-SEM is highly favored by many researchers because it permits the estimation of complex models with numerous constructs, indicators, and paths by researchers without requiring distributional assumptions of the data [66,67].

The data analysis approach employed was a two-way technique consisting of the measurement model and the structural model, as suggested by Stegmann [68], Lomax [69], and Chin [70]. To ascertain any potential relationships between latent items and observable items, the measurement model was run. On the other hand, the purpose of the structural model was to ascertain how the independent and dependent variables relate to one another.

7. Results

7.1. Measurement Model Assessment

The measurement model was thoroughly evaluated to ensure its accuracy and dependability by measuring convergent validity and discriminant validity. The degree of correlation between a construct's indicators, which indicates that they measure the same concept, was used to assess convergent validity. Discriminant validity was examined to ensure that the constructs were unique and not unduly related to one another. These assessments guarantee that the theoretical constructs and their relationships are accurately reflected in the measuring model.

7.1.1. Convergent Validity

The assessment of the level of agreement among indicators of the same construct is known as convergent validity. Cronbach's alpha and composite reliability (CR), as suggested by Hair et al. [71], were taken into consideration in order to establish convergent validity. According to Henseler et al. [72], a concept must be above the acceptable threshold of 0.7 for both Cronbach's alpha and composite reliability in order to meet the convergent validity criterion. The data shown in Table 3 clearly show that all constructs have composite reliability and Cronbach's alpha (α) values greater than 0.7.

Table 3. Convergent validity.

Variable	Cognitive Load	Engagement	Motivation	VIF	CA	CR
C1	0.619	0.430	0.493	1.377	0.851	0.855
C2	0.695	0.504	0.564	1.671		
C3	0.691	0.495	0.529	1.629		
C4	0.723	0.594	0.599	1.580		
C5	0.699	0.509	0.507	1.627		
C6	0.768	0.545	0.588	2.025		
C7	0.669	0.503	0.446	1.487		
C8	0.732	0.585	0.601	1.781		
E1	0.544	0.710	0.499	1.521	0.840	0.843
E2	0.587	0.754	0.587	1.687		
E3	0.611	0.807	0.581	1.978		
E4	0.528	0.704	0.504	1.496		
E5	0.561	0.764	0.503	1.793		
E6	0.505	0.730	0.524	1.660		
M1	0.525	0.532	0.686	1.538	0.827	0.839
M2	0.483	0.499	0.718	1.615		
M3	0.598	0.582	0.768	1.781		
M4	0.569	0.526	0.689	1.510		
M5	0.469	0.419	0.611	1.446		
M6	0.513	0.435	0.630	1.411		
M7	0.434	0.311	0.519	1.340		
M8	0.581	0.507	0.748	1.860		

Notes: C = Cognitive load; E = Engagement; M = Motivation; VIF = Variance inflation factor; CA = Cronbach's alpha; CR = Composite Reliability.

7.1.2. Discriminant Validity

Discriminant validity, conversely, evaluates the uniqueness of dissimilar constructs. The tighter heterotrait–monotrait (HTMT) ratio [73] and cross loading of indicators [70,72] were employed to evaluate the discriminant validity. The loadings of each indicator must exceed all of its cross loadings in order for the discriminant validity criteria to be met; additionally, HTMT ratios of approximately 0.9 are advised [73,74]. It is evident from Table 3 that the item load for each construct is larger on that particular construct than on other constructions. Table 4 further shows that, with an HTMT of 0.9, all values are roughly the threshold value as recommended by Henseler et al. [73] and Ab Hamid et al. [74].

Table 4. Heterotrait–monotrait ratios (HTMT).

Relationship	Heterotrait–Monotrait Ratio (HTMT)
Engagement <-> Cognitive Load	0.878
Motivation <-> Cognitive Load	0.921
Motivation <-> Engagement	0.848

The multicollinearity of the constructs was examined by examining the variance inflation factor (VIF) values in order to further guarantee discriminant validity. Regression analyses use the statistical metric known as VIF to assess the level of multicollinearity. Multicollinearity is the state in which there is a significant degree of correlation between two or more independent variables in a regression model. When analyzing the model findings, multicollinearity might make it difficult to discern the distinct effects of the independent variables on the dependent variable [75]. For multicollinearity not to exist, Kock [75] suggests that VIF values be less than 3.3. According to Table 3, every variable's VIF satisfies the required threshold, showing no problem with multicollinearity

7.2. Structural Model Assessment

The structural model was assessed after validating the measurements model. The bootstrap resampling method was used, which entailed selecting 5000 sub-samples with replacements from the initial sample of 387 in order to determine the significance of the path coefficients in the structural model. The explanatory power of the structural model was assessed by looking at its ability to predict endogenous constructs using the coefficient of determination (R2). The findings of the structural model assessment are shown in Figure 5. The model fit is also measured by the standardized root mean square residual (SRMR). The model's SRMR value was 0.063, indicating a good fit.

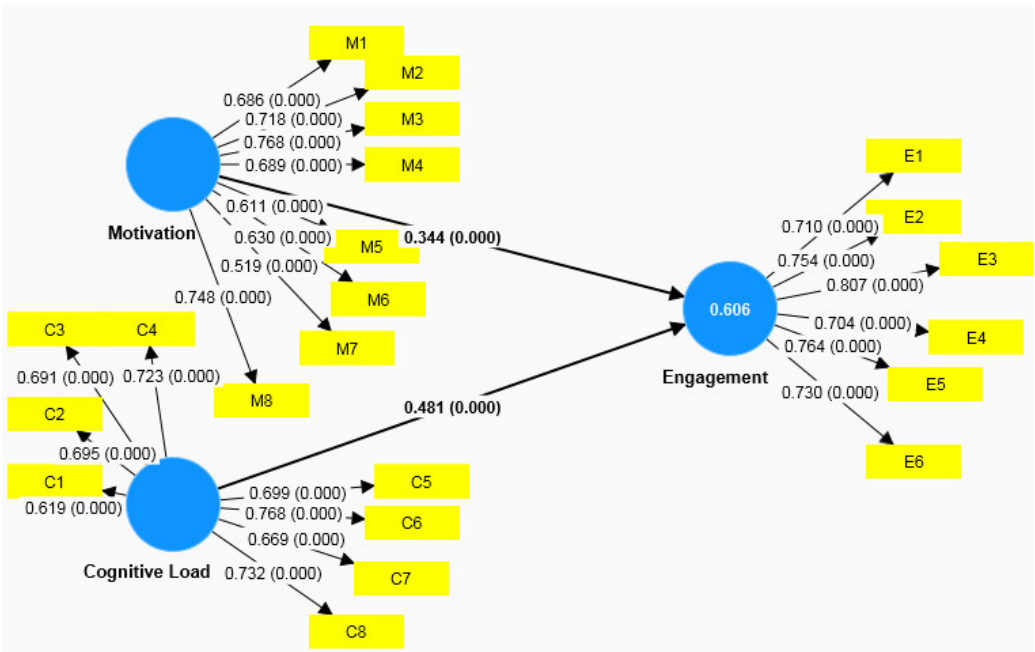

Figure 5. Path analysis of the research model.

As indicated in Table 5, motivation (M) significantly influenced engagement (E) in learning (β = 0.344, p = 0.000), confirming H1.

Table 5. Correlation path testing results.

Path	Path Coefficient	*p*-Values	Result
M->E	0.344	0.000 ***	Significant
C->E	0.481	0.000 ***	Significant

Note: M = Motivation E = Engagement C = Cognitive load *** $p < 0.00$.

Additionally, it was found that engagement (E) in learning was strongly influenced by cognitive load (C) ($\beta = 0.481$, $p = 0.000$), confirming H2 as well. The model's predictive ability was ascertained by looking at the coefficient of determination, or R2. Figure 5 demonstrates how motivation and cognitive load affected learning engagement, with R2 = 0.606. Therefore, a significant 60.6% of the variance in learning engagement was explained by the two independent variables of motivation and cognitive load. Figure 5 also shows the factor loading of each indicator, that is, the degree to which each indicator and the variable it measures are related. As depicted on the diagram, all the indicators show a strong relationship with their respective variables (greater than 0.5), illustrating the extent to which the observable indicator explains the variable (in this case, motivation and cognitive load).

8. Discussion

This study aimed to investigate how gamification impacts students' motivation to learn and their cognitive load to enhance learning engagement.

Data received from 387 students were used to test the model. One construct from the integration of two motivation theories (ARCS and SDT) and one construct from CLT were used to form the model.

The research model explained 60.6% of the change in engagement in learning. This finding suggests that the research model efficiently predicted students' enhancement in learning engagement through the use of gamification. Both determinants (Motivation and Cognitive load) influenced engagement in learning, with cognitive load having a greater impact on engagement. This implies that students are of the opinion that gamification supports them in reducing the mental effort of studying and also helps them to memorize concepts taught more than inciting or motivating them to study. The findings in this study agree with previous studies [2,11,38], which found that through gamification, motivation positively influences learners' engagement in learning. Furthermore, this study concurs with previous studies such as Manzano-León et al. [45] and Darejeh's [42] that gamification reduces cognitive burden and assists students with concept memorization to improve academic performance. However, while this study found that cognitive load has a bigger impact on student's engagement in learning than motivation, some previous studies have found the opposite [76].

These findings have significant implications for educational policies and strategies.

Firstly, the focus on cognitive advantages, like less mental strain and enhanced memory retention, indicates that incorporating gamified components into the curriculum should be given priority in educational initiatives. By doing this, teachers can design cognitively accessible and deeper understanding, thus promoting learning environments. This change may result in a more individualized approach to education, with the goal of making learning less intellectually demanding and more intuitive.

Secondly, these results may have an impact on policies pertaining to the development of curricula and methods of instruction in higher education. Educational establishments could be persuaded to provide teachers with the necessary training to successfully integrate gamification into their lesson plans. This might entail creating materials, instruments, and systems that make it easier to introduce gaming aspects into conventional classroom environments, resulting in a more dynamic and interesting learning environment.

Furthermore, higher education authorities should fund projects that investigate creative approaches to gamify educational materials in a variety of subject areas. Policy may be

put in place to encourage the adoption of gamification in a variety of educational situations, schools, and universities by recognizing the cognitive benefits of gamification.

Lastly, a more thorough examination of the function of motivation in education may result from this realization. Motivation is still important, but instead of depending only on outside stimulants, educators may develop learning experiences that inherently lower cognitive load and improve recall. The ease of comprehending and the sense of accomplishment that comes from grasping topics could encourage a more sustained type of student engagement. As a result, these findings may prompt a reassessment of the educational strategies and policies governing education today, encouraging the use of gamification as a way to enhance cognitive processes during the learning process and, eventually, provide more engaging and successful educational experiences.

9. Conclusions

Empirical studies have demonstrated that learner engagement is significantly impacted by cognitive load and motivation, which are both influenced by gamification. It has been discovered that, of these variables, cognitive load has a stronger relationship with engagement levels. Comprehensive testing proved the validity and reliability of the model that was utilized to produce these results.

The findings mean that adding game aspects to lesson plans could be beneficial for lecturers since it could lessen cognitive burden. Students' motivation to learn can be enhanced by reducing the mental demands placed on them, which will raise their levels of engagement in higher education environments.

By incorporating fun into the learning process, gamification increases students' interest in learning [77]. This preference for gamified learning settings suggests that when learning is fun and engaging, students are more likely to stay motivated and engaged. As a result, adding gamification to higher educational environments promotes a more effective and engaging learning environment, in addition to making learning more desirable.

Limitations and Future Research

In this study, the motivation and cognitive load were measured in the short term, that is, one semester, which might not adequately convey the gamification's long-term effects. Also, because cognitive load might be subjective, the task performances, such as quizzes, used to measure it might not give a clear picture of it. In addition, the study's wider applicability is limited by the context-sensitivity of gamification, which means that what works in one environment could not be as effective in another. Furthermore, the study was limited to a sample of 407 undergraduate students. Future studies should increase the sample size and include postgraduate students to make it easier to generalize the findings to broader populations. Moreover, a future study can compare the impact of gamification on STEM and non-STEM programs to enable stakeholders to gain a better understanding of how gamification needs to be adjusted for various educational settings. This comparison helps educators create more effective, customized gamified learning experiences for a range of learning needs by offering insightful information about its varying effects on motivation, cognitive load, and learning outcomes.

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Conflicts of Interest: The authors declare no conflict of interest.

Appendix A

Demographics						
	B. Gender		**C. Course Schedule**		**D. Level of Study**	
	Male	[ ]	Day (Regular)	[ ]	Level 100 (1st year)	[ ]
	Female	[ ]	Weekend	[ ]	Level 200 (2nd year)	[ ]
	Not willing to disclose	[ ]			Level 300 (3rd year)	[ ]
					Level 400 (4th year)	[ ]
Part 1—Impact of technology-enhanced game elements (gamification) on Motivation						
1.1	The multiple chances I was given to carry out activities boosted my confidence to perform those activities. *Strongly Disagree [ ], Disagree [ ], Neither disagree nor agree [ ], Agree [ ], Strongly Agree [ ]*					
1.2	Because I was aware that when I performed well in an activity, I would get a point or badge, I considered the activities relevant and hence spent more time studying. *Strongly Disagree [ ], Disagree [ ], Neither disagree nor agree [ ], Agree [ ], Strongly Agree [ ]*					
1.3	I found the course satisfying, considering the crossword puzzle and the games, which were used as quizzes in the course. *Strongly Disagree [ ], Disagree [ ], Neither disagree nor agree [ ], Agree [ ], Strongly Agree [ ]*					
1.4	The feedback on my performance, be it from the lecturer or course mates, resulted in productive interactions with the course participants. *Strongly Disagree [ ], Disagree [ ], Neither disagree nor agree [ ], Agree [ ], Strongly Agree [ ]*					
1.5	Being given the opportunity to present myself in a particular way as an avatar gave me the confidence to learn. *Strongly Disagree [ ], Disagree [ ], Undecided [ ], Agree [ ], Strongly Agree [ ]*					
1.6	As I carry out some activities, I get a sense of satisfaction when images and congratulatory messages pop up on the screen to cheer me on. *Strongly Disagree [ ], Disagree [ ], Undecided [ ], Agree [ ], Strongly Agree [ ]*					
1.7	My attention is captured when I see a countdown timer in a quiz. *Strongly Disagree [ ], Disagree [ ], Undecided [ ], Agree [ ], Strongly Agree [ ]*					
1.8	Presenting the course like a treasure hunt incited me to learn. *Strongly Disagree [ ], Disagree [ ], Undecided [ ], Agree [ ], Strongly Agree [ ]*					
Part 2—Impact of technology-enhanced game elements on Cognitive Load						
2.1	As I was given further chances to attempt tasks, less mental effort was required to understand the task or content. *Strongly Disagree [ ], Disagree [ ], Undecided [ ], Agree [ ], Strongly Agree [ ]*					
2.2	The point or badge I received after completing a task successfully stimulated me to study. *Strongly Disagree [ ], Disagree [ ], Undecided [ ], Agree [ ], Strongly Agree [ ]*					
2.3	Knowing that high achievers in our course will be published for all my course mates to see made me to be more focused on my studies. *Strongly Disagree [ ], Disagree [ ], Undecided [ ], Agree [ ], Strongly Agree [ ]*					
2.4	Since some of the quizzes administered to us were like familiar games, I found the design of quizzes very convenient for learning. *Strongly Disagree [ ], Disagree [ ], Undecided [ ], Agree [ ], Strongly Agree [ ]*					
2.5	Since I was given an immediate response on how I performed in an activity, my mind was settled and not distracted. *Strongly Disagree [ ], Disagree [ ], Undecided [ ], Agree [ ], Strongly Agree [ ]*					
2.6	The humorous animated objects intermittently showing up after the successful completion of a task or an activity in the course mentally influenced me to study well. *Strongly Disagree [ ], Disagree [ ], Undecided [ ], Agree [ ], Strongly Agree [ ]*					

2.7	The timers set in the quizzes mentally alerted me to take the quizzes. *Strongly Disagree [ ], Disagree [ ], Undecided [ ], Agree [ ], Strongly Agree [ ]*
2.8	The presentation of the course as a story involving some challenges, heightened my alertness in the course. *Strongly Disagree [ ], Disagree [ ], Undecided [ ], Agree [ ], Strongly Agree [ ]*
	Part 3—Impact on engagement after taking the gamified course
3.1	*Affective*: I felt that what we were learning in the course was important, and that motivated me to study. *Strongly Disagree [ ], Disagree [ ], Neither disagree nor agree [ ], Agree [ ], Strongly Agree [ ]*
3.2	*Affective*: I found the way the course was delivered captivating. *Strongly Disagree [ ], Disagree [ ], Neither disagree nor agree [ ], Agree [ ], Strongly Agree [ ]*
3.3	*Behavioral:* I always look forward to attending lectures because the activities in the lectures are interesting. *Strongly Disagree [ ], Disagree [ ], Neither disagree nor agree [ ], Agree [ ], Strongly Agree [ ]*
3.4	*Behavioral:* When I run into a difficult task, I keep working at it until I have solved it because of the satisfaction I get after solving the task. *Strongly Disagree [ ], Disagree [ ], Neither disagree nor agree [ ], Agree [ ], Strongly Agree [ ]*
3.5	*Cognitive:* I put in every effort to see the similarities and differences between things I was learning in the course and things I already know. *Strongly Disagree [ ], Disagree [ ], Neither disagree nor agree [ ], Agree [ ], Strongly Agree [ ]*
3.6	*Cognitive:* I tried to match what I already knew with things I was trying to learn in the course. *Strongly Disagree [ ], Disagree [ ], Neither disagree nor agree [ ], Agree [ ], Strongly Agree [ ]*

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